

**Negative Exponents with Variable Bases**Question Bank 3.T2 **ANSWER KEY**

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Show your work. Circle your final answer.

**For questions 1 through 6, write an equivalent expression using only positive exponents. Assume no variable equals 0.**

1.  $a^{-6}$

$$\frac{1}{a^6}$$

2.  $\frac{1}{k^{-3}}$

$$k^3$$

3.  $7y^{-4}$

$$\frac{7}{y^4}$$

The 7 has no negative exponent; it stays in the numerator.

4.  $x^{-2}y^5$

$$\frac{y^5}{x^2}$$

5.  $\frac{m^4}{n^{-3}}$

$$m^4n^3$$

6.  $\frac{2p^{-5}q}{r^{-2}}$

$$\frac{2qr^2}{p^5}$$

$p^{-5}$  moves down;  $r^{-2}$  moves up; 2 and  $q$  stay.

For questions 7 and 8, apply the laws of exponents. Write each answer using only positive exponents.

7. Write an equivalent expression.

a.  $x^3 \cdot x^{-8}$

$$x^{-5} = \frac{1}{x^5}$$

b.  $(3y^{-2})^3$

$$27y^{-6} = \frac{27}{y^6}$$

c.  $(2a^{-3}b^2)(5a^7b^{-6})$

$$10a^4b^{-4} = \frac{10a^4}{b^4}$$

8. Write an equivalent expression.

a.  $\frac{m^2}{m^9}$

$$m^{-7} = \frac{1}{m^7}$$

b.  $\frac{12x^3y^{-2}}{4x^5y}$

$$3x^{-2}y^{-3} = \frac{3}{x^2y^3}$$

$$3 - 5 = -2; -2 - 1 = -3.$$

c.  $\left(\frac{a^3}{4b^{-7}}\right)^5$

$$\frac{a^{15}}{4^5b^{-35}} = \frac{a^{15}b^{35}}{1,024}$$

**Working backwards.**

9. Find the value of  $n$  that makes each statement true.

a.  $x^{-5} \cdot x^n = x^2$

$$n = 7$$

b.  $\frac{y^n}{y^4} = \frac{1}{y^6}$

$$n = -2$$

$$n - 4 = -6.$$

c.  $(m^n)^3 = \frac{1}{m^{12}}$

$$n = -4$$

$$3n = -12.$$

10. Let  $x = 3$ .

a. Find the value of  $x^2$  and the value of  $x^{-2}$ .

**9 and 1/9**

b. Find the value of  $-x^2$ . Is it equal to  $x^{-2}$ ? Explain.

**-9. No —  $x^{-2}$  is the reciprocal of  $x^2$ , which is 1/9;  $-x^2$  is the opposite of  $x^2$ .**

11. Which expression is equivalent to  $5x^{-3}$ ?

☒ A.  $\frac{5}{x^3}$

☐ B.  $\frac{1}{5x^3}$

☐ C.  $-5x^3$

☐ D.  $\frac{1}{125x^3}$

**A**

12. Select ALL of the expressions that are equivalent to  $\frac{1}{x^6}$ .

☒ A.  $x^{-6}$

☒ B.  $x^2 \cdot x^{-8}$

☒ C.  $\frac{x^3}{x^9}$

☒ D.  $(x^{-2})^3$

☐ E.  $-x^6$

☐ F.  $(x^{-3})^{-2}$

**A, B, C and D**